ME0522 · HYDRAULICS & PNEUMATICS

Simplified Performance-Based Hydraulic Excavator Design

A 30-ton excavator hydraulic system, designed from external performance targets inward.

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BENCHMARK Caterpillar 330 · 350 bar class



AGENDA: a· 10 MINUTES Story!

A Designer's Path, From Requirements to Hardware

01 **Scope & Design Approach**
The excavator as a hydraulic manipulator

02 **Design Philosophy & Benchmark**
Cat 330 performance targets

03 **Simplified Architecture**
Pump → valve → actuators

04 **Mechanical Loads & Forces**
From digging forces to bore sizes

05 **Component Selection**
Pump, motor, cylinders, valve

06 **Losses & Benchmark Comparison**
Validation after pressure losses

Scope & Design Approach

An excavator converts fluid power into linear and rotary motion — a hydraulic manipulator. We size only the core power path, starting from required forces.

IN SCOPE

The hydraulic power path:

- Variable-displacement **pump** and load-sensing **multiway valve**
- **Travel motors** driving the crawler tracks
- **Boom, stick & bucket** double-acting cylinders
- Boundary: **pump outlet → actuator return line**

EXCLUDED FROM DETAILED SIZING

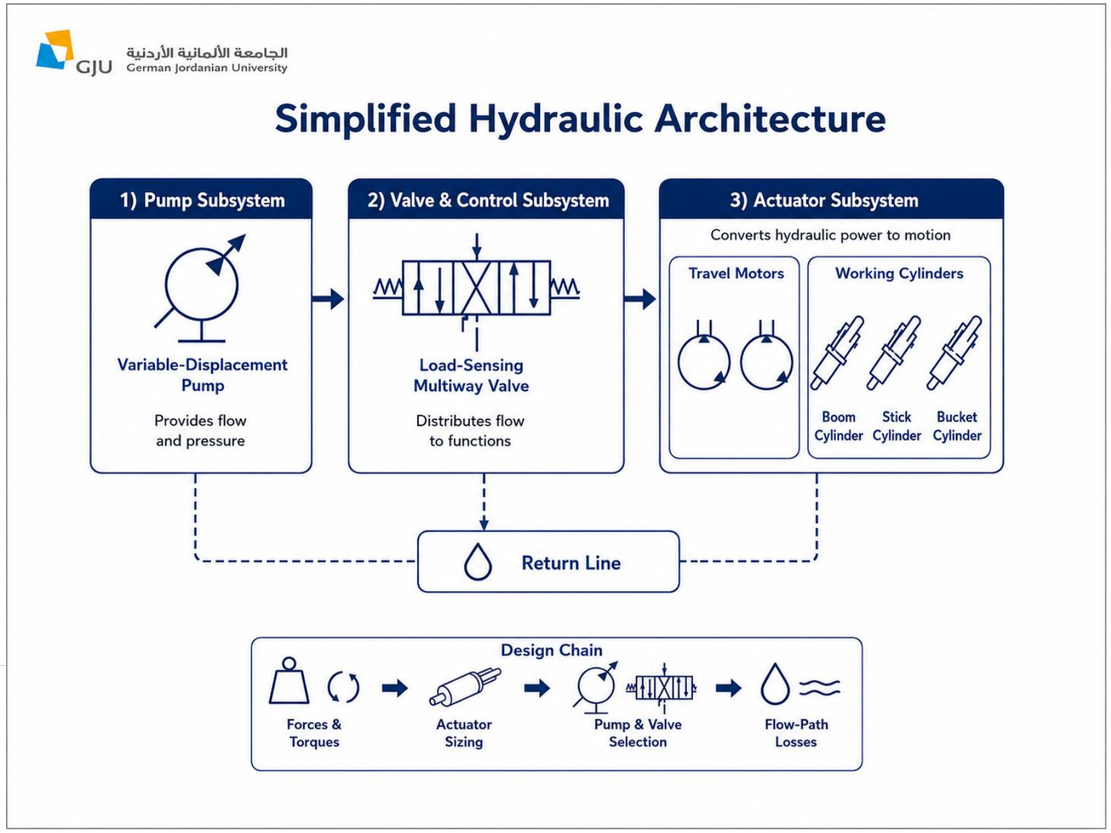
Out of scope

Engine and prime mover sizing

Swing drive system

Tank, cooler & filtration

Exact proprietary linkage geometry



Design Philosophy & Benchmark Targets

A **performance-based** approach: Cat 330, but meeting its operating requirements. Its published figures become our design inputs.

248 kN	179 kN	145 kN	350 bar
Drawbar pull (travel force target)	Bucket digging force target	Stick digging force target	System pressure class adopted



Benchmark machine: **Caterpillar 330**, 30-ton class — also targeting ~560 L/min main flow and ~204 kW engine power.

Mechanical Loads on Cylinders and Motor

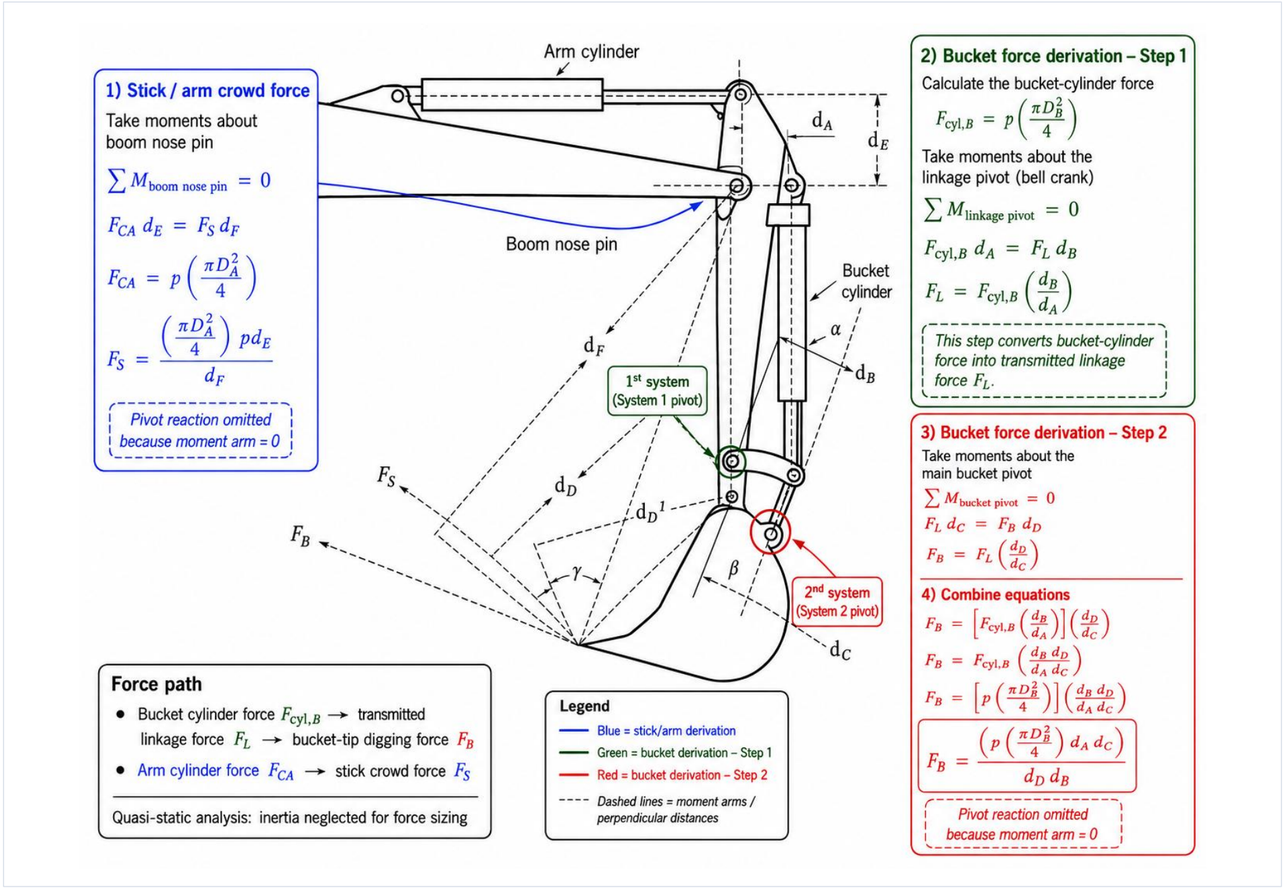


Fig. — Linkage moment-equilibrium: cylinder force → bucket-tip force

Travel — drawbar to motor torque

$F_{track} = F_{drawbar}/2 = 124\ kN \rightarrow T_{motor} \approx 1200\ N{\cdot}m\ /\ side$

Implements — force through linkage

$F_{cyl} = p \cdot (\pi D^2/4)\ F_{tip} = k_L \cdot F_{cyl},\ k_L \approx 0.35$

≈ 511 kN

Required bucket-cylinder force

≈ 414 kN

Required stick-cylinder force

Motor Section: Bosch Rexroth A6VM Series 71, NG280

Operating Data

Selection item	Value
Required torque	$\approx$ **1200 Nm**
Size	NG 280
Type	Bent-axis axial piston variable motor
Displacement	280 cm^3/rev
Nominal pressure	450
Maximum pressure	500
Rated torque at 350 bar	1404Nm

Toque Validation

From Data Sheet:

$$T_m = \frac{V_g \Delta p \eta_{hm}}{20\pi}$$
$$T_m = 1404$$
$$T_{req} \approx 1200$$
$$T_m > T_{reg}$$

Flow Demand, a problem!

$$Q = V_d n$$

At the estimated motor speed of 1600 rpm:
$Q \approx 448$ L/min per motor!!
two motors **= 896L/min**

A solution

The high flow value is **not** treated as a **continuous operating point**.

The A6VM is a variable-displacement motor. As shown by the HA2 characteristic curve, motor displacement changes with working pressure:

- High pressure / high load → high displacement for traction torque
- Lower load / higher travel speed → reduced displacement to limit flow demand

Therefore, maximum drawbar pull and maximum travel speed are not assumed at the same time.

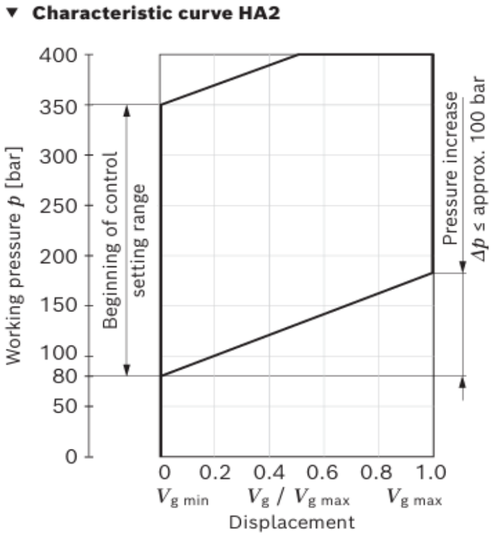


Figure 1: HA2 Variable-Displacement Control Curve

Cylinder Section: Boom, Stick & Bucket Actuators

Benchmark Targets

- Bucket = 179 kN
- Stick = 145 kN

Linkage	Equivalent factor	Approximate values	Result
Bucket	$k_{L,B} = \frac{d_A d_C}{d_D d_B}$	$d_A = 155, d_B = 400, d_C = 1500, d_D = 1660$	0.35
Stick	$k_{L,S} = \frac{d_E}{d_F}$	$d_E = 1120, d_F = 3200$	0.35

Cylinder	Type	Initial checked bore	Final selected bore	Estimated velocity
Boom	Double-acting hydraulic cylinder	140 mm	140 mm	$\approx \mathbf{0.15}$ **m/s**
Stick	Double-acting hydraulic cylinder	130 mm	150 mm	$\approx \mathbf{0.26}$ **m/s**
Bucket	Double-acting hydraulic cylinder	140 mm	145 mm	$\approx \mathbf{0.28}$ **m/s**

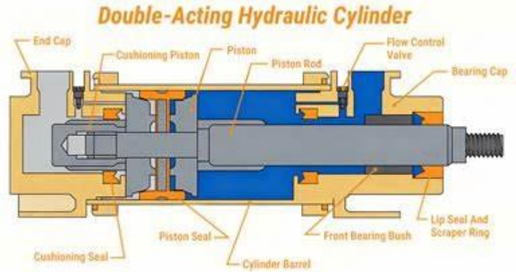


Figure 2: Double acting Hydraulic Cylinders

Pump Section: Axial piston, variable · load-sensing (DRS0)

Operating Data

Selection item	Value
Pump type	Variable axial piston
Displacement, Vg	145 cm³/rev
Design speed, n	2000 rpm
Volumetric efficiency, η_v	0.95
Nominal pressure	350 bar
Flow per pump	276 L/min
Total design flow	552 L/min

LS Control Margin

- means the pump pressure must be **slightly higher than the actuator/load pressure** so oil can actually move through the control valve.
- it appears as a **pressure drop across the valve/control metering path.**
- ΔpLS ≈ 14 bar.
- The **14 bar** is therefore a **control pressure margin**.
  It is not pipe friction, but it is pressure reserved for valve control, so it is not fully available at the actuator

Pump Pressure and Flow Curve

The curve shows the pump operating envelope:

- Flow increases from qv min to qv max
- Pressure is limited near 350 bar
- At high pressure, the pump cannot stay at maximum flow continuously
- Load-sensing control changes displacement according to valve demand

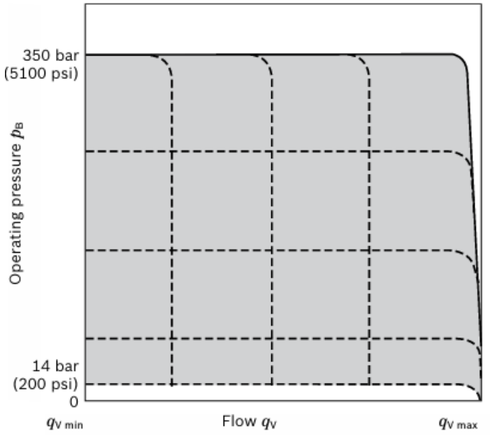


Figure 1: HA2 Variable-Displacement Control Curve

Load Sensing Multiway Valve

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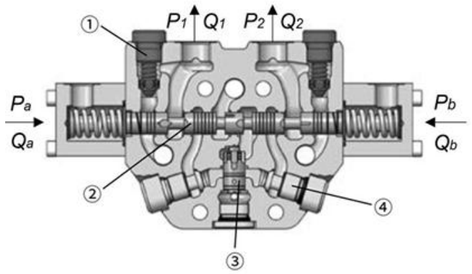


Figure 4: Multiway Valve Demonstration

Full-System Losses & Validation

SUPPLY	– PIPE & FITTINGS	– LS CONTROL MARGIN	= AT ACTUATOR
350 bar	3.1 bar	14 bar	≈ 333 bar
Pump pressure class	Friction over longest branch	Load-sensing reserve	Pressure available for work

ACTUATOR	@ 350 BAR	@ 333 BAR	REQUIREMENT	RESULT
Travel motor — A6VM NG280	1404 N·m	1336 N·m	≈ 1200 N·m	**Suitable**
Boom cylinder — Ø140 mm	539 kN	513 kN	benchmark	**Suitable**
Stick cylinder — Ø150 mm	217 kN	206 kN	145 kN	**Suitable**
Bucket cylinder — Ø145 mm	203 kN	193 kN	179 kN	**Suitable**

APPENDIX A · COST

Bill of Materials

ITEM	QTY	ESTIMATED TOTAL
Main pump — A11VLO 145	2	$8,000
Travel motor — A6VM NG280	2	$6,000
Planetary final drive	2	$6,000
Boom / stick / bucket cylinders	4	$5,400
Load-sensing multiway valve block	1	$1,500
Hoses, fittings & safety valves	1 set	$2,000

ESTIMATED MAIN HYDRAULIC BOM

$28.9k ++

Order-of-magnitude estimate from current supplier listings — realistic for a 30-ton machine.

Benchmark Comparison & Conclusion

DESIGN ITEM	BENCHMARK / TARGET	FINAL SELECTED	OUTCOME
System pressure	350 bar	350 bar	Same class
Pump flow	560 L/min	552 L/min	Within 1.4%
Travel motor torque	≈ 1200 N·m	1336 N·m	Met after losses
Boom / stick / bucket bore	140 / 150 / 135	140 / 150 / 145	Matches benchmark
Protection & damping	Industry features	Suspension + cushioning	Added for realism

Starting only from external performance targets, the design lands a coherent, **simplified hydraulic system** suitable as a preliminary design for a 30-ton excavator.